

ICDFA NETWORK SECURITY FUNDAMENTALS

PRACTICAL LABORATORY 2

TCP IP Ethernet ARP and Wireshark Analysis

WEEK	TCP/IP ARP and Wireshark Analysis
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Registration No	C11/26/FCDF/17
GUIDED LAB WINDOW	25-September,2026
SELF-PACED WINDOW	Monday- Friday after the guided session
SUBMISSION DEADLINE	2-October,2026, 23:59 WAT
LMS ITEMS	Lab 2
Linux Version	6.18. 12+kali-amd64
Vm hostname	kali

Purpose of this laboratory

This laboratory teaches you to observe local and routed communication instead of relying only on ping results. You will capture authorised traffic on the Client, identify Ethernet and ARP fields, follow ICMP, DNS and TCP exchanges, and explain why the destination MAC address can identify the gateway while the destination IP address identifies a remote server.

Begin from the LAB-BASELINE-VALIDATED snapshots created in Lab 1. Do not change VirtualBox adapter mappings during this exercise.

Learning outcomes

- Select the correct Wireshark capture interface using Linux evidence.
- Identify ARP requests, ARP replies and the learned neighbour entry.
- Interpret Ethernet source, destination and EtherType fields.

- Compare local destination MAC addresses with end-to-end IPv4 addresses.
- Identify ICMP, DNS and TCP traffic with display filters.
- Save and label a packet capture as assessment evidence.

Authorisation and safety rules

- Perform every action only inside the ICDFA VirtualBox lab supplied for this course.
- Keep ICDFA-LAN, ICDFA-DMZ and ICDFA-TRANSIT as Internal Networks. Never bridge them to a school, workplace or public network.
- Do not scan, capture or test systems that are not listed in this guide.
- Take the required snapshot before changing firewall, addressing, DHCP or routing settings.
- Do not expose passwords, private keys or unrelated personal information in screenshots or packet captures.
- Stop and ask the instructor if a command would affect the host computer or a network outside the virtual lab.

ICDFA virtual lab topology

The lab contains four virtual machines and four logically separate network segments. Traffic between the LAN, DMZ and upstream network must pass through the intended routing and firewall components.

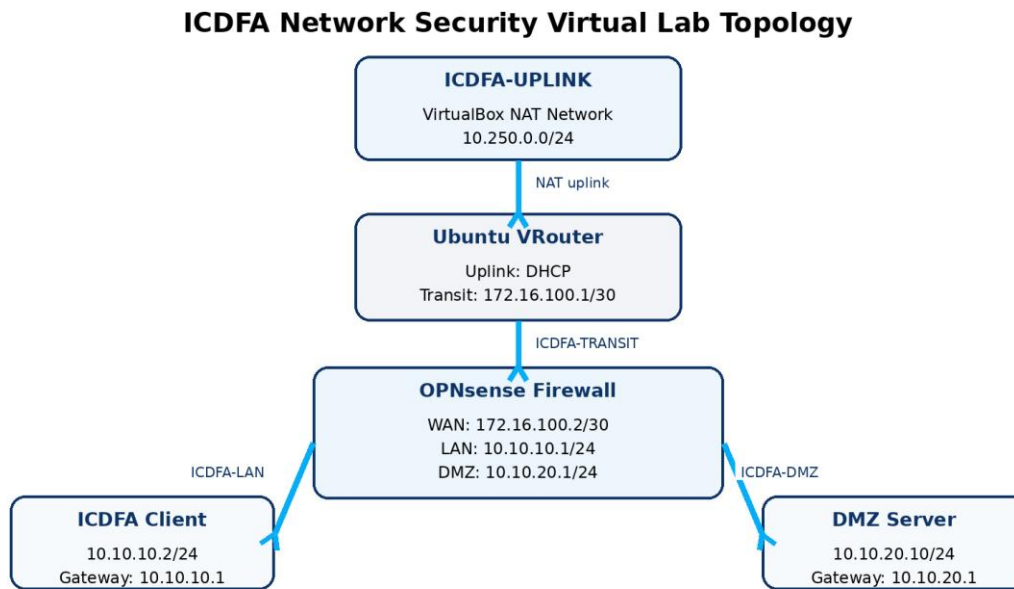


Figure 1 ICDFA Network Security virtual lab topology

Required starting state

- All four VMs are using the validated network configuration from Lab 1.
- The Client has 10.10.10.2/24 and can reach OPNsense at 10.10.10.1.
- The DMZ Server has 10.10.20.10/24 and its Nginx and Bind9 services are active.
- Wireshark 4.x is installed on the Client and the student account belongs to the wireshark group.

Part 1 Establish a clean baseline

```
ip -4 -br address
ip -br link
ip route
ip neigh show
groups
wireshark --version | head -n 1
```

1. Identify the Client interface that has 10.10.10.2/24. Recorded its name as enp0s3.
2. Recorded the hexadecimal MAC address shown for enp0s3 .
3. Confirmed that the default route points to 10.10.10.1 through enp0s3.
4. Confirmed that Wireshark appears in the groups output.

Item	Student observation
Client interface name	enp0s3
Client IPv4 address	10.10.10.2/24
Client MAC address	08:00:27:35:d0:7a
Default gateway	10.10.10.1
Existing neighbour entry for 10.10.10.1	52:54:00:12:35:00
Wireshark version	Wireshark 4.2.2

Part 2 Start a capture on the correct interface

5. Opened Wireshark from the Ubuntu applications menu.
6. Located enp0s3. The activity graph beside the interface should respond when traffic is generated.
7. Double-clicked enp0s3 to start capturing. Precaution was: Do not choose lo, docker0 or an inactive adapter.
8. Leave the capture running and open a second terminal window.

Part 3 Generated a fresh gateway ARP exchange

A host does not issue a new ARP request when it already has a usable neighbour entry. Delete only the gateway entry so that the Client must resolve it again. Substitute the interface discovered in Part 1.

```
sudo ip neigh del 10.10.10.1 dev enp0s3 2>/dev/null || true
ping -c 4 10.10.10.1
```

Return to Wireshark and click the red Stop button after the four ping replies complete.

Part 4 Analysed the ARP request

9. Enter arp in the Wireshark display-filter bar and press Enter.
10. Select the packet whose Info column asks Who has 10.10.10.1? Tell 10.10.10.2.
11. Expand Ethernet II and record the destination and source MAC addresses.
12. Expand Address Resolution Protocol and record the opcode, sender MAC, sender IP and target IP.
13. Apply arp.opcode == 1 and confirm that the selected request remains visible.

ARP request field	Expected observation	Student value
Ethernet destination	ff:ff:ff:ff:ff:ff	PCSSystemtec_4b:3c:58
Ethernet source	Client MAC	PCSSystemtec_35:d0:7a
ARP opcode	1 request	1
Sender IPv4	10.10.10.2	10.10.10.102
Target IPv4	10.10.10.1	10.10.10.1
Target MAC	Unknown or all zeroes in the request	08:00:27:4b:3c:58

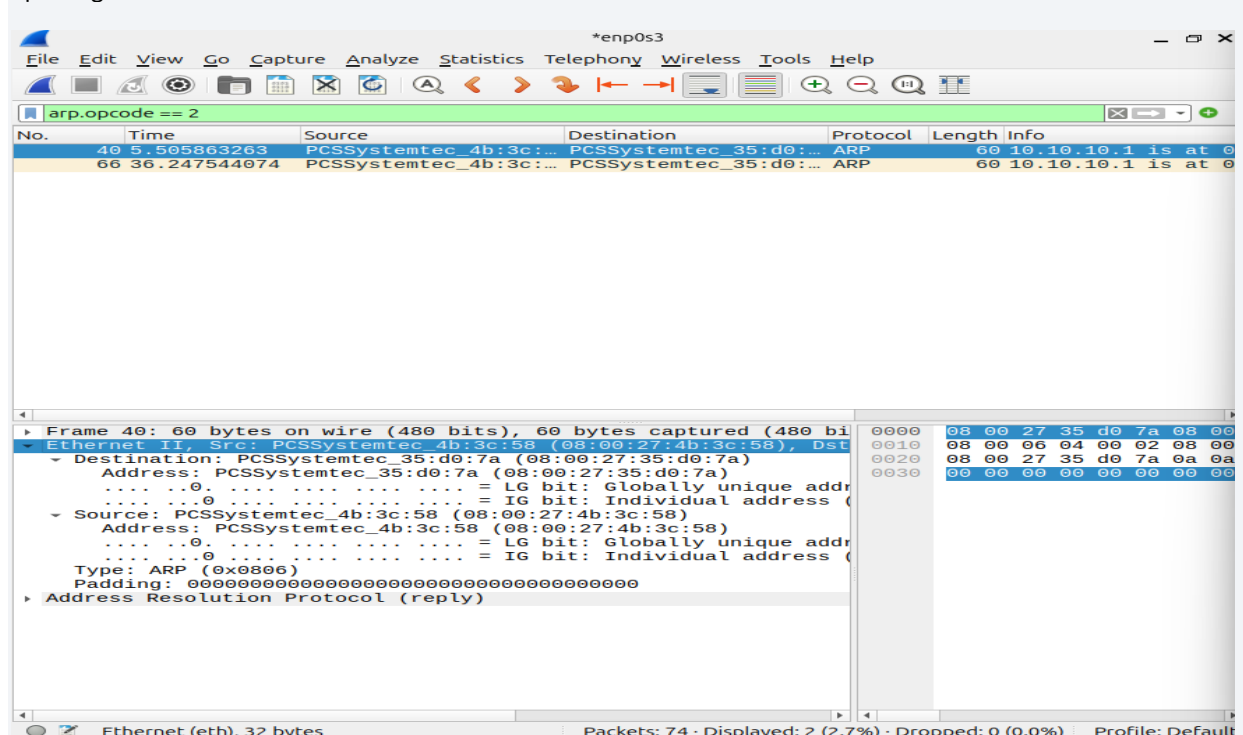
Wireshark interface showing an ARP request packet (No. 39) selected. The display filter is `arp.opcode == 1`. The packet list shows two ARP packets. The packet details pane shows the structure of the ARP request: Ethernet II (Src: PCSSystemtec_35:d0:7a, Dest: PCSSystemtec_4b:3c:58) and Address Resolution Protocol (Opcode: 1, Sender IP: 10.10.10.2, Target IP: 10.10.10.1, Sender MAC: 08:00:27:4b:3c:58, Target MAC: 08:00:00:00:00:00). The packet bytes pane shows the raw data in hexadecimal and ASCII.

Part 5 Analysed the ARP reply

14. Apply arp.opcode == 2.
15. Select the reply that states 10.10.10.1 is at a MAC address.
16. Expand Ethernet II and ARP. Record the reply source MAC, destination MAC and sender IPv4 address.
17. Compare the reply source MAC with the OPNsense LAN MAC recorded in Lab 1.

ARP reply field	Expected observation	Student value
Ethernet destination	Client MAC	08:00:27:35:d0:7a
Ethernet source	OPNsense LAN MAC	08:00:27:4b:3c:58
ARP opcode	2 reply	2
Sender IPv4	10.10.10.1	10.10.10.1
Target IPv4	10.10.10.2	10.10.10.102

```
ip neigh show 10.10.10.1
```



Part 6 Compare local and routed delivery

18. Start a new capture on enp0s3.
19. In a terminal, run ping -c 4 10.10.20.10.
20. Stop the capture and apply icmp.
21. Select the first ICMP echo request. Expand Ethernet II and Internet Protocol Version 4.
22. Record the Ethernet destination MAC and IPv4 destination address.

Layer	Destination observed	Meaning
Ethernet	OPNsense LAN MAC	The next device on the local ICDFA-LAN segment
IPv4	10.10.20.10	The final DMZ Server destination

Expected result: The Ethernet destination is the gateway MAC, not the DMZ Server MAC. The IPv4 destination remains 10.10.20.10.

Part 7 Observe DNS and TCP

Start one final capture on enp0s3, then run the following commands.

```
dig @10.10.20.10 dmz.icdfa.test
curl -I http://10.10.20.10
```

Stop the capture after the HTTP response appears. Apply each display filter separately.

Display filter	What to identify
dns	DNS query for dmz.icdfa.test and the answer 10.10.20.10
udp.port == 53 tcp.port == 53	DNS transport traffic
tcp.flags.syn == 1 && tcp.flags.ack == 0	The Client's first TCP SYN
tcp.flags.syn == 1 && tcp.flags.ack == 1	The Server's SYN ACK
tcp.port == 80	The complete HTTP TCP conversation
ip.addr == 10.10.10.2 && ip.addr == 10.10.20.10	Only the Client and DMZ Server IPv4 conversation

Display filters hide packets from view; they do not delete them from the capture file.

Part 8 Save the packet evidence

23. In Wireshark, select File and then Save As.
24. Save the final capture as StudentID_Lab02_Evidence.pcapng.
25. Store it in the folder specified by the instructor.
26. Open the saved file once and confirm that the packets and filters are still available.

Troubleshooting guide

Problem	Reason	Action
No interface appears	Wireshark group membership or capture permissions are missing	Confirm groups output, then log out and back in after membership is corrected.
Packets appear only on lo	The loopback interface was selected	Stop and select the interface carrying 10.10.10.2.

Problem	Reason	Action
No ARP request appears	The neighbour entry was already cached	Delete only the 10.10.10.1 entry and repeat the ping while capturing.
ARP requests repeat without a reply	Local attachment, subnet or gateway interface is wrong	Check ICDFA-LAN, cable state, Client /24 address and OPNsense LAN.
ARP succeeds but DMZ ping fails	Routing or firewall policy is preventing the flow	Check the Client route, OPNsense route and firewall log.
DNS filter shows no answer	Bind9, DNS policy or query target is wrong	Verify bind9 and query 10.10.20.10 directly.

CONCLUSION:

In this laboratory exercise, deep packet analysis was successfully conducted using Wireshark to investigate Layer 2 and Layer 3 traffic dynamics within the ICDFA virtual environment. The analysis confirmed that local host discovery relies on Layer 2 Ethernet broadcast requests (ff:ff:ff:ff:ff:ff) paired with direct unicast replies, while inter-subnet traffic to the DMZ (10.10.20.10) retains the remote target IP at Layer 3 but uses the default gateway's MAC address (08:00:27:4b:3c:58) at Layer 2 for local delivery. All captured network traffic flows were successfully recorded, filtered, and exported to Haseeb_Lab02_Evidence.pcapng for assessment compliance and forensic verification.